

Python intro interview questions

1. What is the difference between a compiler and an interpreter?

Answer:

A compiler translates the entire source code into machine code or bytecode before execution, producing an executable file. An interpreter, on the other hand, translates source code into machine code or byte code line by line at runtime, executing.

2. How does an interpreter execute code written in a high-level programming language?

Answer:

An interpreter reads the source code line by line, translates each line into machine code or byte code, and executes it immediately. It does not produce an intermediate executable but directly executes the code at runtime.

3. Which programming languages typically use compilers for code execution?

Answer:

Languages like C, C++, Java, and Rust typically use compilers for code execution. These languages are compiled into machine code before execution.

4. Which programming languages typically use interpreters for code execution?

Answer:

Languages like Python, JavaScript, Ruby, and PHP typically use interpreters for code execution. These languages are executed line by line at runtime without the need for a separate compilation step.

5. How does the choice between compilation and interpretation affect code execution speed?

Answer:

Generally, compiled languages tend to have faster code execution because the compilation process optimizes code beforehand, and the resulting machine code is executed directly. In contrast, interpreted languages may have slower execution since each line of code is translated and executed at runtime.

6.What is binary code, and how is it related to low-level programming languages?

Answer:

Binary code is the representation of instructions and data in a binary (base-2) numeral system, consisting of only two digits: 0 and 1. In computing, binary code is used to represent all data and instructions that a computer processes, including machine instructions, memory addresses, and data values.

Low-level programming languages, such as Assembly language, directly correspond to binary code. Each instruction in a low-level language is represented by a specific pattern of binary digits, which the computer's processor can execute directly.

7.What is Programming?

Answer:

Programming is the process of creating a set of instructions that tell a computer how to perform a task.

Programming can be done using a variety of computer programming-languages such as JavaScript, Python, and C++.

8.What is Coding?

Answer:

Coding is a list of step by step instructions that get computers to do what you want them to do.

Coding makes it possible for us to create computer software, games, apps and websites.